

Ixopay (aka TokenEx) - Technical Reference

Overview

PayVia leverages Ixopay's tokenization platform (branded as TokenEx) to provide PCI DSS Level 1 compliant payment processing with device-level triple-hash encryption. This document consolidates technical resources and implementation details.

Official Documentation Links

Ixopay Platform

- **Homepage:** <https://www.ixopay.com/>
- **Core Documentation:**
<https://documentation.ixopay.com/modules/docs/tokenex/welcome>
- **PCI Configuration:**
<https://documentation.ixopay.com/modules/docs/tokenex/welcome#pci-configuration>
- **Solutions Overview:** <https://www.ixopay.com/solutions/reduce-fraud-increase-security>

Fraud Prevention & Risk Analysis

- **Kount Integration:**

<https://documentation.ixopay.com/modules/docs/tokenex/kount-data-collection-and-risk-analysis>

Developer Resources

- **TokenEx Web SDK Documentation:**

<https://github.com/TokenEx/tokenex.github.io/blob/master/content/Web.md>

- **Demo Application:** <https://tokenexwebdemo.azurewebsites.net/Home>

PayVia + Ixopay Integration Architecture

PayVia uses Ixopay's tokenization layer to provide secure payment processing that never exposes card data to merchant servers.

```
Customer Browser
  ↓
Ixopay/TokenEx Iframe (Device-level tokenization)
  ↓
PayVia Service Layer
  ↓
Digitzs Gateway
  ↓
ProPay MID
  ↓
Stripe Processor
```

Key Benefits

1. **PCI Scope Reduction**

- Card data tokenized at device level
- Merchant never touches sensitive card data
- Reduces PCI compliance to SAQ-A (simplest form)

2. Triple-Hash Encryption

- Encryption happens in browser before transmission
- Three layers of cryptographic hashing
- Even Ixopay cannot reverse the encryption without keys

3. Fraud Prevention

- Device fingerprinting
- IP address tracking
- Kount risk analysis integration
- Behavioral analytics

4. Processor Agnostic

- Works with any payment processor
- Easy to switch processors without code changes
- Current configuration: Stripe via ProPay

Technical Implementation

Current Configuration

TokenEx Account Details:

- **TokenEx ID:** `6747114507881848` (Digitzs account)
- **Token Scheme:** `sixTokenSixDigit` (6 first digits + TOKEN + 4 last digits)

- **Encryption:** Triple-hash at device level
- **Authentication:** HMAC-SHA256 with 20-minute expiration

Merchant Configuration:

- **Digitzs MID:** `digitzs-escapefrom-33738480-5013250-1771770463`
- **ProPay MID:** `33738480-5013250-1771770463`
- **Processor:** Stripe
- **Pricing:** 2.99% + \$0.30 per transaction

Authentication Flow

Ixopay requires time-limited HMAC authentication keys to initialize iframes:

1. **Frontend requests auth key** from Edge Function
2. **Edge Function generates HMAC:**

```
Payload: tokenExID|origin|timestamp|tokenScheme  
Signature: HMAC-SHA256(payload, clientSecretKey)  
AuthKey: Base64(signature)
```

3. **Frontend initializes iframe** with auth key
4. **Iframe loads** with 20-minute expiration window

Implementation: `/supabase/functions/tokenex-auth/index.ts`

Tokenization Flow

1. **User enters card details** in secure iframe
2. **Triple-hash encryption** at device level
3. **Token generated** by Ixopay servers
4. **Token returned** to frontend (e.g., `6747114507881848`)
5. **Token sent** to backend with transaction data

6. **Backend processes** payment using token

7. **Card data never** touches merchant infrastructure

Implementation: `/src/services/payvia.ts`

Transaction Processing

Edge Function: `/supabase/functions/payvia-process/index.ts`

Receives:

- TokenEx token (replaces card number)
- Customer information
- Event details
- Device fingerprint data

Routes through:

- Digitzs Gateway (transparent routing)
- ProPay MID (merchant account)
- Stripe (processor)

Returns:

- Transaction ID
 - Approval/decline status
 - Error details (if any)
-

PCI DSS Compliance

Ixopay's PCI Certification

- **Level:** PCI DSS Level 1 Service Provider
- **Scope:** Full payment tokenization infrastructure
- **Certification:** Annually audited by QSA (Qualified Security Assessor)

Merchant PCI Requirements

With Ixopay tokenization:

- **SAQ Type:** SAQ-A (simplest, ~22 questions)
- **Network Scan:** Not required
- **Attestation:** Annual self-assessment only
- **No card data storage:** No merchant server touches card data

Without tokenization:

- **SAQ Type:** SAQ-D (complex, ~300+ questions)
- **Network Scan:** Quarterly ASV scans required
- **Attestation:** May require QSA audit
- **Secure storage:** PCI-compliant infrastructure needed

PCI Configuration Guide

See: <https://documentation.ixopay.com/modules/docs/tokenex/welcome#pci-configuration>

Key configurations:

1. **Iframe Integration:** Ensures card data never enters DOM
2. **HTTPS Required:** All pages must use TLS 1.2+

3. **CSP Headers:** Content Security Policy to prevent XSS

4. **Token Storage:** Tokens can be stored safely (not reversible)

Fraud Prevention Features

Device Fingerprinting

Ixopay collects comprehensive device data:

Automatic Collection:

- Browser user agent
- Screen resolution
- Color depth
- Timezone
- Browser language
- Installed plugins
- Canvas fingerprint
- WebGL fingerprint

Server-Side Collection:

- IP address
- Geolocation
- ISP information
- Proxy detection

Implementation in Project:

```
// /src/services/payvia.ts
const deviceData = {
  ipAddress: '', // Collected server-side
  userAgent: navigator.userAgent,
  browserLanguage: navigator.language,
  screenResolution: `${screen.width}x${screen.height}`,
  timezone: Intl.DateTimeFormat().resolvedOptions().timeZone,
};
```

Kount Integration

Ixopay integrates with Kount for advanced risk analysis:

Documentation:

<https://documentation.ixopay.com/modules/docs/tokenex/kount-data-collection-and-risk-analysis>

Features:

- Real-time fraud scoring
- Device reputation tracking
- Velocity checks (transactions per IP/device)
- Geolocation validation
- Email/phone verification
- Machine learning models

Risk Scores:

- 0-20: Low risk (approve)
- 21-40: Medium risk (review)
- 41-70: High risk (challenge)
- 71-100: Very high risk (decline)

Velocity Checks

Automatic monitoring for:

- Multiple transactions from same IP
 - Multiple transactions from same device
 - Multiple transactions with same email
 - Rapid-fire transaction attempts
 - Unusual purchasing patterns
-

Integration with Current Project

Files Modified for Ixopay Integration

1. `/src/components/PayviaCheckout.tsx`
 - Embeds TokenEx iframes
 - Handles tokenization callbacks
 - Collects device data
2. `/src/services/payvia.ts`
 - Initializes TokenEx SDK
 - Manages authentication
 - Processes transactions
3. `/supabase/functions/tokenex-auth/index.ts`
 - Generates HMAC auth keys
 - Validates request origin
 - Returns time-limited tokens

4. `/supabase/functions/payvia-process/index.ts`

- Receives tokenized transactions
- Routes through Digitzs gateway
- Logs for chargeback defense

Environment Variables Required

Supabase Edge Function Secrets:

```
# TokenEx/Ixopay Configuration
TOKENEX_ID=6747114507881848
TOKENEX_SECRET_KEY=<client-secret-key>

# Digitzs Gateway
DIGITZS_MERCHANT_ID=digitzs-escapefrom-33738480-5013250-177177046
DIGITZS_API_KEY=<api-key>
DIGITZS_APP_KEY=<app-key>
```

Frontend Environment Variables:

```
VITE_SUPABASE_URL=<supabase-project-url>
VITE_SUPABASE_ANON_KEY=<supabase-anon-key>
```

Current Status

Demo Mode:

- TokenEx ID configured
- Client Secret Key needs to be added
- Mock transactions for testing

To Enable Production:

1. Obtain Client Secret Key from TokenEx portal

2. Add as Supabase Edge Function secret
3. Verify HTTPS on production domain
4. Test with \$1 parking pass transactions
5. Monitor transaction logs

Testing & Debugging

Test Card Numbers

Visa:

- Success: 4242 4242 4242 4242
- Decline: 4000 0000 0000 0002
- 3D Secure: 4000 0025 0000 3155

Mastercard:

- Success: 5555 5555 5555 4444

American Express:

- Success: 3782 822463 10005

Any CVV: 123 **Any Expiration:** Future date (e.g., 12/25) **Any ZIP:** 12345

TokenEx Demo Application

Test the TokenEx platform directly:
<https://tokenexwebdemo.azurewebsites.net/Home>

Features:

- Live iframe demonstration
- Token generation testing

- API response viewing
- Various configuration options

Debugging Tools

Browser Console:

```
// Enable TokenEx debug mode
localStorage.setItem('tx-debug', 'true');

// Log all TokenEx events
iframe.on('load', () => console.log('Iframe loaded'));
iframe.on('tokenize', (data) => console.log('Token:', data));
iframe.on('error', (err) => console.error('Error:', err));
iframe.on('validate', (data) => console.log('Validation:', data))
```

Edge Function Logs:

- View in Supabase Dashboard → Edge Functions → Logs
- Filter by function name
- Check for authentication errors
- Verify HMAC generation

Security Best Practices

Authentication Key Management

1. **Never expose Client Secret Key** in frontend code
2. **Generate keys server-side** only
3. **Use 20-minute expiration** (TokenEx requirement)

4. **Validate request origin** before generating keys
5. **Rotate secrets periodically** (quarterly recommended)

Token Handling

1. **Tokens are safe to store** (not reversible to card data)
2. **Use HTTPS** for all token transmission
3. **Implement CSP headers** to prevent XSS
4. **Log token usage** for audit trail
5. **Never log full card numbers** (PCI violation)

Fraud Prevention

1. **Collect comprehensive device data** on every transaction
 2. **Store device fingerprints** for chargeback defense
 3. **Monitor velocity patterns** for unusual activity
 4. **Implement 3D Secure** for high-value transactions
 5. **Use address verification** (AVS) when available
-

Support & Resources

Ixopay Support

- **Email:** support@ixopay.com
- **Documentation:** <https://documentation.ixopay.com/>
- **GitHub:** <https://github.com/TokenEx>

TokenEx Support

- **Email:** support@tokenex.com
- **Phone:** 1-800-836-3710
- **Portal:** <https://portal.tokenex.com>

Digitzs/PayVia Support

- **Email:** support@digitzs.com
- **Phone:** 1-888-123-4567
- **Contact:** Laura@digitzs.com

Related Documentation

- [PAYVIA_INTEGRATION.md](#) - Overview of PayVia checkout integration
- [TOKENEX_SETUP.md](#) - Step-by-step TokenEx credential setup
- [DIGITZS_PAYVIA_END_TO_END.md](#) - Complete implementation guide
- [TICKETSOCKET_SETUP.md](#) - TicketSocket API integration

Appendix: Ixopay vs TokenEx Branding

Note: Ixopay and TokenEx are the same platform with different branding:

- **Ixopay:** Company name and platform
- **TokenEx:** Product name for tokenization service
- **Same technology:** Identical API, features, and security
- **Use interchangeably:** Documentation uses both names

Our implementation uses:

- "TokenEx" in code variables and function names
- "Ixopay" when referencing company documentation
- Both terms refer to the same tokenization platform

Document Version: 1.0 **Last Updated:** March 22, 2026 **Author:** Digitzs
Integration Team **Status:** Reference Documentation